Schlumberger
Carbon Services

# Risk Management Status

## Illinois Basin – Decatur Project

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## Table of Contents


Executive Summary ..... 3
INTRODUCTION ..... 4
RISK WORKSHOPS ..... 4
RISK WORKSHOP OUTCOMES ..... 5
RISK MANAGEMENT ..... 5
PRE-INJECTION EVALUATION ..... 5
MID-PROJECT RISK RE-EVALUATION ..... 6
APPENDIX A: WORKSHOP ROSTERS AND WORKING GROUPS ..... 7
First Workshop / First Cohort: February 2008 ..... 7
Second Workshop / Second Cohort: March 2008 ..... 8
Abbreviations in workshop rosters ..... 9
APPENDIX B: SCENARIOS ..... 10
APPENDIX C: RISK TREATMENTS (RISK RESPONSE ACTIONS) AND DOCUMENTATION SUMMARY ..... 15
APPENDIX D: "RACI" CHART ..... 26

# Risk Management Status

## Executive Summary

Illinois Basin – Decatur Project (IBDP) is a project undertaken by the Midwest Geological Sequestration Consortium (MGSC), whose members include Illinois State Geologic Survey (ISGS), Archer Daniels Midland Corporation (ADM), Schlumberger Carbon Services, and others. It is a "Phase III large-scale demonstration project" of the Regional Carbon Sequestration Program funded by National Energy Technology Laboratory, U.S. Department of Energy. Risk management for IBDP began with project kickoff in 2008, continues presently, and will carry on for the duration of the project. Through December 2011, the stages of risk management emphasized risk identification, evaluation, treatment development, and treatment execution tracking; near-term future emphasis is placed on risk re-evaluation in light of early monitoring data to be acquired in mid-2012.

During two workshops held in early 2008, more than 40 professionals most knowledgeable in project facts, uncertainties, and plans evaluated 123 concepts called "FEPs" – Features, Events, and Processes – for their associated risks to project goals and values. The defined values include safety, research objectives, and public acceptance as well as technical aspects of successfully storing and monitoring injected CO2. In six working groups, the experts quantified Severity and Likelihood of negative impact (whose product, S*L, is defined as "Risk") according to five-point categorical scales. Evaluations were made by both group consensus and independent scoring; the independent scores prevented the potential obscuration of valid "worst-case" concepts that could occur through averaging of group scores or through group dynamics.

The higher-risk FEPs (roughly the top half of the evaluated list) were used to develop approximately 100 concrete scenarios that would result – if they occurred – in negative impact. The scenarios were used in turn to develop >200 risk treatments (risk response actions / "RRAs") involving well engineering, subsurface characterization, computer modeling, monitoring, public communications, and other efforts. Risk treatments are actions that reduce the severity and/or the likelihood of negative impact.

Risk treatments were grouped into 29 areas of function or expertise, called Risk Response Action Groups (RRAGs). Each RRAG was assigned to a specific individual responsible for organizing and documenting the completion of the actions. Documentation was maintained on an electronic spreadsheet developed by the risk manager and accessible 24/7 to all individuals involved in risk treatments.

Early in the project, half of the 206 RRAs were identified as "required to be completed before injection start", unless their incomplete status were deemed acceptable by the IBDP Principal Investigator (PI). When the injection permit was received from Illinois EPA, about 25 of those actions were not documented as complete. Their status was reviewed by the Schlumberger Carbon Services Risk Manager and Project Manager, and the IBDP PI. Before authorizing compressor start, the Schlumberger Carbon Services PM and IBDP PI agreed that the outstanding items did not pose immediate threats to safety, were adequately mitigated through other actions, or were no longer pertinent; and that accordingly overall risk stood at an acceptable level. The considerable development of knowledge about the project's subsurface details, engineered systems, and human systems, during the two years after the RRA list was generated, established confidence that risk levels were acceptably low.

# INTRODUCTION

Illinois Basin – Decatur Project (IBDP) is a project undertaken by the Midwest Geological Sequestration Consortium (MGSC). MGSC members include Illinois State Geologic Survey (ISGS), Archer Daniels Midland Corporation (ADM), Schlumberger Carbon Services, Trimeric Corporation, State geological surveys in Kentucky and Indiana, and academic and research institutions across the United States. IBDP is a "Phase III large-scale demonstration project" of the Regional Carbon Sequestration Program funded by National Energy Technology Laboratory, U.S. Department of Energy. The central purpose of IBDP is to safely inject, into a suitable subsurface geologic reservoir, one million tons of carbon dioxide ($CO_2$) over a period of three years, and to monitor its fate and transport during this period and subsequent years.

Risk management for IBDP began with the project kickoff meeting in 2008, continues now, and will carry on for the duration of the project. Through December 2011, the stages of risk management emphasized risk identification and evaluation; developing risk treatments (risk response actions); executing and tracking risk treatments. In early to mid-2012 (i.e. at the time this report is written), risk management emphasis is placed on risk re-evaluation in light of (1) Early monitoring data being acquired in mid-2012; and (2) Operational experience.

Detailed descriptions of risk assessment processes and early results are given in the two papers listed below. Copies of both papers may be obtained online, or from ISGS or Schlumberger Carbon Services. These risk assessment processes have been described in several semi-public meetings, chiefly under the auspices of International Energy Agency, Greenhouse Gas Program (IEAGHG), but in other semi-public forums as well.

- Hnottavange-Telleen, Ken, Ethan Chabora, Robert J. Finley, Sallie E. Greenberg, Scott Marsteller, Risk management in a large-scale CO2 geosequestration pilot project, Illinois, USA 10th Intl Conference on Greenhouse Gas Control Technologies, Amsterdam (Elsevier)
- Hnottavange-Telleen, K., I. Krapac, C. Vivalda, 2008, "Illinois Basin – Decatur Project: initial risk-assessment results and framework for evaluating site performance." 9th Intl Conference on Greenhouse Gas Control Technologies, Nov. 16-21, 2008, Washington D.C., USA.

# RISK WORKSHOPS

During two workshops held in February and March 2008, more than 40 professionals most knowledgeable in project facts, uncertainties, and plans evaluated 123 concepts called "FEPs" – Features, Events, and Processes – for their associated risks to project goals and values. A list of 178 FEPs pertinent to $CO_2$ Capture and Sequestration (CCS) projects has been developed by Quintessa Ltd. and published by IEAGHG; the FEPs list used for IBDP was a slightly refined version of the Quintessa list. FEPs are relatively broad concepts; their evaluation does not require identifying specific chains of events (scenarios).

To evaluate risk, it is mandatory to define the specific entities or values that may be *at risk.* For IBDP, the defined project values include safety, research objectives, and public acceptance, in addition to technical aspects of successfully storing and monitoring injected $CO_2$.

Organized into six topical working groups (three of which met at each of the two workshops) (Appendix A), the experts quantified Severity and Likelihood of negative impact according to five-point categorical scales.

The product of Severity and Likelihood, S*L, is defined as "Risk", and the quantified magnitude of "risk" can therefore vary along a scale from 1 to 25. This calculated "risk" value is the main comparator or ranking value of the evaluated entities.

Evaluations were made both by group consensus and by follow-up independent scoring. Generally, averaged group scores produce the most consistent ranking of risked entities. Nonetheless, independent scoring is also done to counter potential "groupthink" effects, and to prevent potentially obscuring valid "worst-case" concepts.

# RISK WORKSHOP OUTCOMES

After all FEPs were evaluated and ranked in terms of Risk = S*L, the higher-risk FEPs (roughly the top half of the evaluated list) were used to develop >200 concrete scenarios (Appendix B) that would result – if they occurred – in negative impact to one or more project values. In contrast to FEPs, scenarios are specific and well defined chains of events. Only scenarios that have plausibly significant impact are addressed. In turn, the higher-risk scenarios were used to develop >200 risk treatments (risk response actions / "RRAs") that involve well engineering, subsurface characterization, computer modeling, monitoring, public communications, and other efforts.

# RISK MANAGEMENT

The set of risk treatments was subdivided among 29 areas of function or expertise. The set of scenarios was also uniquely subdivided among the same 29 areas (Appendix B). Each "risk response action group" (RRAG) is assigned to a specific individual who is responsible for organizing and documenting the completion of defined risk treatments, and is also responsible more generally for managing the risk associated with a set of scenarios. Other individuals are "Accountable for" the defined risk treatments, or are to be "Consulted with" or "Informed", as shown in a "RACI" chart (Appendix D). Documentation is maintained on an electronic spreadsheet developed by the risk manager and accessible 24/7 to all individuals involved in executing, documenting, and verifying risk treatments.

Risk treatments (Appendix C) are actions that reduce the severity and/or the likelihood of negative impact. In standard terminology, reductions in *likelihood* are referred to as "preventions", while reductions in *severity* are called "mitigations". For a given level of effort, preventions are normally preferred because they reduce risk for all risk receptors (project values), while mitigations typically reduce risk only for specific receptors. In general, mitigations are applied where sufficient prevention is impossible (e.g. for uncontrollable environmental variables such as severe weather) or impractical. Risk treatments vary in complexity: some are relatively simple actions that can be unambiguously determined as complete, while others entail developing detailed plans that either (a) Will be used during normal project execution, or (b) Will only be used contingently, if a specific situation occurs.

# PRE-INJECTION EVALUATION

Early in the project, approximately 85 of the 200 RRAs were identified as "required for completion before injection start", unless their incomplete status were to be deemed acceptable by the IBDP Principal Investigator (PI). As progress was made toward the start of injection, many of these actions were

completed, were documented as complete by their respective "Responsible" parties, and were verified as complete by their respective "Accountable" parties.

At the moment when the injection permit was received from Illinois EPA, about 25 of the defined risk treatments remained not documented as complete (some actions were known to have been completed, yet were not documented as such). Given the desire to commence injection as soon as practical, the status of actions not documented as complete was reviewed by the Schlumberger Carbon Services Risk Manager and Project Manager and the IBDP PI. The Schlumberger Carbon Services PM and IBDP PI agreed that the risks addressed by the outstanding response actions did not pose immediate threats to safety and/or were adequately addressed through other actions. Consequently, the IBDP PI determined that residual risk stood at an overall acceptable level, and authorized compressor start. The considerable development of knowledge about the project's subsurface details, engineered systems, and human systems, during the two years after the RRA list was generated, established confidence that risk levels were acceptably low.

# MID-PROJECT RISK RE-EVALUATION

By mid-2012, significant operational experience will have been gained, and significant additional project data will be available. The new data will consist of several rounds of sampling from the in-reservoir observation well, continuous pressure and temperature data at multiple surface points and depth levels, and processed 3D seismic data from which $CO_2$ plume distribution will be interpreted. All of this information will form the basis for re-evaluating risk at a greater level of knowledge than existed at project start.

In general terms, the objectives of mid-project re-evaluation are the same as those in the early-project evaluation: to identify and evaluate sources of risk, to plan actions that will reduce those risks that are unacceptably high, and to initiate a management structure that will maintain risk at acceptable levels and will document that this is the case. The key difference from the initial evaluation is the evolution of the project from the initiation-creation stage toward a progressive development phase, with an eye toward beginning the establishment and documentation of the conditions that will eventually enable project closure. While the overall project values will remain the same, the evolution of project activities and shorter-term objectives may call for revisiting the scales of potential negative impact.

# APPENDIX A: WORKSHOP ROSTERS AND WORKING GROUPS

### First Workshop / First Cohort: February 2008

| First | Last | Organization | Role or Specialty | 1st Cohort Group | also in 2nd Cohort |
|---|---|---|---|---|---|
| Lee | Cunningham | ADM | Legal Counsel | AIR | |
| Dwight | Peters | Carbon* | SCS manager | AIR | |
| Scott | Frailey | ISGS | MGSC Assist Deputy Program Dir | AIR | * |
| Sallie | Greenberg | ISGS | Public Outreach | AIR | * |
| Zhining | Tao | ISWS | Atmospheric modeling | AIR | |
| Jennifer | Lewicki | LBNL | CO2 field techniques/modeling | AIR | |
| Arthur | Wells | NETL | Tracers | AIR | |
| Ken | Hnottavange-Telleen | Carbon* | Risk Assessment / process leader | floater | * |
| Scott | Marsteller | Carbon* | SCS - Project Manager | floater | * |
| Ivan | Krapac | ISGS | Geochemistry/ MMV coordinator | floater | |
| Ron | Peterson | ADM | Operations | SNS | |
| Jim | Kirksey | Carbon* | Petroleum Engineer | SNS | * |
| Tim | Larson | ISGS | Shallow Geophysics | SNS | |
| Ed | Mehnert | ISGS | Hydrology | SNS | * |
| William | Roy | ISGS | Geochemistry/modeling | SNS | |
| Curt | Oldenburg | LBNL | Modeling CO2 in vadose zone | SNS | |
| Brian | Strazisar | NETL | Vadose zone geochem. | SNS | |
| Tom | Stone | ADM | Manager | SUB | * |
| Alan | Brown | Carbon* | Static Earth Model | SUB | * |
| Marcia | Coueslan | Carbon* | Geophysics-seismic | SUB | |
| Andrew | Duguid | Carbon* | Well Integrity | SUB | |
| Paul | Hughes | Carbon* | Well Construction | SUB | * |
| Uwe | Rinck | Carbon* | Omegalok downhole seismic | SUB | |
| Jeff | Thompson | Carbon* | Seismic | SUB | |
| Rob | Finley | ISGS | MGSC Program Director | SUB | * |
| Rob | Finley | ISGS | Program Director | SUB | |
| Hannes | Leetaru | ISGS | Bedrock Geologist | SUB | * |
| Jim | Johnson | LLNL | Geochem/geomech modeling | SUB | |
| Rick | Chalaturnyk | U Alberta | Geomechanics | SUB | |

## Second Workshop / Second Cohort: March 2008

| First | Last | Organization | Role or Specialty | 2nd Cohort Group | also in 1st Cohort |
|---|---|---|---|---|---|
| Polly | Wise | FG | IL FutureGen; Il Ofc Coal Devel | COM | |
| Anthony | Pleasant | FG/Mattoon | Mattoon FutureGen | COM | |
| Brian | Moody | FG/Tuscola | Tuscola FutureGen | COM | |
| Sallie | Greenberg | ISGS | Public Outreach | COM | * |
| Hannes | Leetaru | ISGS | Bedrock Geologist | COM | * |
| Grant | Bromhal | NETL | CCS regs / protocols development | COM | |
| Sarah | Forbes | WRI | NGO / public interest | COM | |
| Ken | Hnottavange-Telleen | Carbon* | Risk Assessment / process leader | floater | * |
| Claudia | Vivalda | Carbon* | Risk Assessor | floater | |
| Nikita | Chugunov | SDR | Risk Assessor | floater | |
| Mark | Carroll | ADM | Envtl Compliance Mgr | OEN | |
| Jill | Davidson | ADM | Environmental Manager | OEN | |
| Sarah | Wade | AJW | CCS risk methods study | OEN | |
| Alan | Brown | Carbon* | Static Earth Model | OEN | * |
| Patricia | Van Allen | Carbon* | Legal Counsel | OEN | |
| Scott | Anderson | ED | NGO / public interest | OEN | |
| Angela | Griffin | FG/Mattoon | Mattoon FutureGen | OEN | |
| Alice | Bernard | ISGS | Project Finances | OEN | |
| Rob | Finley | ISGS | MGSC Program Director | OEN | * |
| Tom | Stone | ADM | Manager | SEC | * |
| Paul | Hughes | Carbon* | Well Construction | SEC | * |
| Jim | Kirksey | Carbon* | Petroleum Engineer | SEC | * |
| Scott | Marsteller | Carbon* | SCS - Project Manager | SEC | * |
| Scott | Frailey | ISGS | MGSC Asst. Deputy Program Dir. | SEC | * |
| Ed | Mehnert | ISGS | Hydrology | SEC | * |
| George | Peridas | NRDC | CCS science and politics promotion | SEC | |
| Kevin | Fisher | Trimeric | Compr-Dehydr-Pipeline | SEC | |
| Ray | McKaskle | Trimeric | Compr-Dehydr-Pipeline | SEC | |

## Abbreviations in workshop rosters

**Working Groups** (Topical Scope)

AIR – Wellsite area, Air and atmosphere

SNS – Wellsite area, Surface and Near Surface

SUB – Wellsite area, Subsurface

COM – Community

OEN – Organizational and Nonspatial Topics

SEC – Surface Engineering; Capture and Compression Systems

**Organizations**

| | |
|---|---|
| ADM | Archer Daniels Midland Corporation |
| AJW | AJW Group |
| Carbon* | Schlumberger Carbon Services |
| ED | Environmental Defense |
| FG | FutureGen |
| ISGS | Illinois State Geological Survey |
| ISWS | Illinois State Water Survey |
| LBNL | Lawrence Berkeley Laboratory |
| LLNL | Lawrence Livermore Laboratory |
| NETL | National Energy Technology Laboratory |
| NRDC | Natural Resources Defense Council |
| SDR | Schlumberger Doll Research |
| Trimeric | Trimeric Corporation |
| U Alberta | University of Alberta |
| WRI | World Resources Institute |

# APPENDIX B: SCENARIOS

Scenario risks are managed through Risk Response Actions (RRAs). Scenarios and RRAs belong to Risk Response Action Groups (RRAGs).RRAs, Appendix B. RRAG assignments to individual persons, Appendix C.

| Scen ID# | SCENARIO | Assigned **Risk Response Action Group** | RRAG ID |
|---|---|---|---|
| #Ci07 | Prominent accident (especially involving CO2 release) raises questions about general safety of CCS - even if the accident is ordinary industrial or drilling, not related to CCS specifically. | QHSE PLAN: Subsurface and Seismic Ops | **OE01** |
| #Hs12 | Numerous common industrial, drilling, and driving hazards, typically addressed by existing QHSE practices. | QHSE PLAN: Subsurface and Seismic Ops | **OE01** |
| #Hs14 | Personal weather hazards: Cold, heat, wind, lightning, etc. | QHSE PLAN: Subsurface and Seismic Ops | **OE01** |
| #Ci14 | An accident by any member of the team reflects badly upon the project, and/or upon the image of another consortium member. | QHSE PLAN: Surface and Injection Operations | **OE02** |
| #Fn03 | Cost implications of any accident, including emergency response, medical, equipment damage, delay. | QHSE PLAN: Surface and Injection Operations | **OE02** |
| #Hs10 | Unplanned CO2 release in confined space could cause health hazard to employees working on compressor and other surface facilities. | QHSE PLAN: Surface and Injection Operations | **OE02** |
| #Op06 | Numerous minor traffic accidents occur at or around the site resulting in ADM to lose some confidence in project management and requiring a safety shutdown to hold additional training. | QHSE PLAN: Surface and Injection Operations | **OE02** |
| #Op07 | After injection ceases and the pre closure monitoring continues there will be a shift of focus on HSE that existed during the active injection period, resulting in an accident or unplanned event. | QHSE PLAN: Surface and Injection Operations | **OE02** |
| #Hs11 | Moving vehicle strikes person or damages equipment; traffic hazards onsite or offsite. | SITE SECURITY PLAN | **OE03** |
| #Op01 | Heavy rail traffic results in derailment blocking access to site. | SITE SECURITY PLAN | **OE03** |
| #Sf02 | Persons enter main project site or remote monitoring site and vandalize or steal project equipment. | SITE SECURITY PLAN | **OE03** |
| #Mg02 | The second verification well is drilled after injection has begun. While drilling the well encounters CO2 above the Eau Claire resulting in a well control issue. Project is suspended until leakage path is identified. | DRILLING PLAN | **OE04** |
| #Op08 | Injection well drilling encounters unexpected high pressures requiring well-killing operation resulting in lost days, increased drilling cost, and/or lost measurement opportunities. | DRILLING PLAN | **OE04** |
| #Op12 | Injection well drilling encounters borehole stability problems requiring changes to casing and/or mud program, resulting in lost days, increased drilling cost, and/or lost measurement opportunities. | DRILLING PLAN | **OE04** |
| #Mg07 | CO2 migrates across well seal (cement/casing) at high enough rate that injection is suspended, preventing achievement of project goals. | COMPLETION PLAN: In-Zone Wells | **OE05** |

| Scen ID# | SCENARIO | Assigned **Risk Response Action Group** | RRAG id |
|---|---|---|---|
| #Op04 | Cement dissolution results in very small amounts of CO2 gas present in intermediate/injection string annulus after two years of injection causing remedial work to be performed to finish project. | COMPLETION PLAN: In-Zone Wells | **OE05** |
| #Op15 | Intermittent or poorly controlled injection (temperature, pressure, flow rate, composition) causes rapid degradation of downhole components through chemical or mechanical means. | INJECTION OPERATIONS ENVELOPE | **OE09** |
| #Op16 | Incorrect or inadequate physical process models incorrectly predict response of engineered components, leading to unexpected failure. | INJECTION OPERATIONS ENVELOPE | **OE09** |
| #Ci20 | Action causes failure to comply with permit. | INJECTION OPS AND SHUTIN PLAN | **OE10** |
| #Rs10 | Procurement delay shortens the time available for tests using a critical piece of equipment, causing loss of baseline data about either this equipment or the site. | INJECTION OPS AND SHUTIN PLAN | **OE10** |
| #Op11 | Equipment breakdown causes repair expense and delay. | EQUIPMENT SPARING PLAN | **OE11** |
| #Op13 | Manufacturing of key mechanical components ceases, and insufficient spares have been stocked to meet repair and/or maintenance requirements for the long term, incurring extra cost for custom manufacture. | EQUIPMENT SPARING PLAN | **OE11** |
| #Rs03 | Accident damages custom-manufactured equipment, sensor, tool; replacement or repair delay causes irretrievable information loss during an injection phase. | EQUIPMENT SPARING PLAN | **OE11** |
| #Rs11 | Breakdown of sensor or tool; replacement or repair delay causes irretrievable information loss. | EQUIPMENT SPARING PLAN | **OE11** |
| #Mg03 | While performing yearly monitoring operations due to equipment failure or human error CO2 is allowed to escape from the well uncontrolled for several hours or days. | OPERATIONAL MONITORING PLAN | **OE12** |
| #Sf03 | Major earthquake causes unexpected damage | OPERATIONAL MONITORING PLAN | **OE12** |
| #Fn10 | Unexpected failure requires remediation that is not budgeted | PROGRAMMATIC AND OVERALL RISK | **PM01** |
| #Fn12 | General: change in project scope requires additional funding | PROGRAMMATIC AND OVERALL RISK | **PM01** |
| #Fn13 | DOE fails to deliver promised funding or approve additional funding causing a budget shortfall | PROGRAMMATIC AND OVERALL RISK | **PM01** |
| #Fn14 | Partners fail to provide promised expertise at a reduced rate or total cost share, additional funding required to replace at market rates. | PROGRAMMATIC AND OVERALL RISK | **PM01** |
| #Fn15 | Claim of project-caused damage to geologic resource (water, gas, oil, coal), agriculture, or buildings must be legally defended. If claim is successful, potential additional remedial cost or damage payment. | PROGRAMMATIC AND OVERALL RISK | **PM01** |
| #Zz01 | Any unexpected situation arises for which prep is inadequate, thus impact occurs. | PROGRAMMATIC AND OVERALL RISK | **PM01** |
| #Ci19 | Action causes permit delay | PERMITS | **PM02** |
| #Ci21 | Action causes public opposition which results in permit or project delay. | PERMITS | **PM02** |
| #Fn08 | A permit is modified by government agency to require increased data access and approval by NGOs and/or public; this causes delays leading to price escalation. | PERMITS | **PM02** |

| Scen ID# | SCENARIO | Assigned **Risk Response Action Group** | RRAG id |
|---|---|---|---|
| #Op05 | Lack of local operational experience results in not obtaining a required permit. For biggest media impact, regulators wait to intercede until just before activities begin, resulting in lost days. | PERMITS | **PM02** |
| #Ci17 | Different reservoir models show different result, first mostly in the technical community but then picked up by the media or the general public resulting in loss of confidence in technical competence. | DATA MANAGEMENT PLAN | **PM03** |
| #Op14 | Information technology changes, and archival data are lost due to inability to read older media or formats. | DATA MANAGEMENT PLAN | **PM03** |
| #Ci15 | Cost overruns and failure to meet all goals result in doubt cast on MGSC and SCS ability to manage such projects. | BUDGET | **PM04** |
| #Fn17 | Inaccurate cost estimating and budgeting result in budget shortfall, limiting project activities. | BUDGET | **PM04** |
| #Rs01 | Escalation in cost of research consumables or services reduces the remaining research budget, thus not all goals can be met. | BUDGET | **PM04** |
| #Fn04 | Miscoordination of activities causes resource or timing conflict, causing delay. | STAFF AND COORDINATION | **PM05** |
| #Fn07 | Procurement delay for critical equipment causes increased costs due to rescheduling, and/or overall project delay. | STAFF AND COORDINATION | **PM05** |
| #Fn16 | Poor execution of project activity causes need to re-do. | STAFF AND COORDINATION | **PM05** |
| #Op03 | Unfamilar work flows result in delays impacting budget. Tensions develop among parties leading to a loss of focus on HSE resulting in an accident or unplanned rig event. | STAFF AND COORDINATION | **PM05** |
| #Rs06 | Miscoordination causes loss of measurement opportunity or loss of acquired data. | STAFF AND COORDINATION | **PM05** |
| #Ci02 | Release of project monitoring data that IS NOT of technical concern spurs public concern ... | COMMUNICATIONS POLICY | **PM08** |
| #Ci03 | Release of project monitoring data that IS of technical concern spurs public concern ... | COMMUNICATIONS POLICY | **PM08** |
| #Ci04 | Unreleased project monitoring data that IS of technical concern reaches the public and spurs public concern... | COMMUNICATIONS POLICY | **PM08** |
| #Ci09 | Public knowledge of significant planning error causes loss of public confidence in ability to conduct CCS. | COMMUNICATIONS POLICY | **PM08** |
| #Fn05 | Low perm forces redesign of CO2 delivery system, incurring direct cost and delay. | INJECTIVITY BACKUP PLAN | **PM12** |
| #Fn11 | Low injection rates require a longer injection time frame and therefore more operating funds not budgeted | INJECTIVITY BACKUP PLAN | **PM12** |
| #Rs07 | Failure to find adequate poro/perm zones prevents injecting at desired rate, prevents conducting desired tests, could cause project cancellation. | INJECTIVITY BACKUP PLAN | **PM12** |
| #Rs14 | Overestimation of frac gradient. Mt. Simon is supposed to have a 1 psi/ft frac gradient and all injection modeling is based upon this. If testing shows that frac gradient is actually much lower (0.6) this would have a major impact on injection rates as well as causing the IEPA to rethink injection permit. | INJECTIVITY BACKUP PLAN | **PM12** |
| #Hs02 | CO2 moves via wellbore or fault directly into USDW | NONCONTAINMENT RESPONSE PLAN | **PM13** |

| Scen ID# | SCENARIO | Assigned **Risk Response Action Group** | RRAG id |
|---|---|---|---|
| #Hs05 | Mechanically caused rupture (e.g. equipment collision) in CO2 pipeline or wellhead creates toxic atmosphere or IDLH low-oxygen condition. | NONCONTAINMENT RESPONSE PLAN | **PM13** |
| #Hs06 | Rupture in CO2 pipeline or wellhead allows rapid gas escape, creating mechanical and temperature hazards. | NONCONTAINMENT RESPONSE PLAN | **PM13** |
| #Hs08 | Very large mass of CO2 continuously migrates via faults or fractures across multiple seals to soil or atmosphere, creating high CO2 concentration in a subgrade space or topographic low. | NONCONTAINMENT RESPONSE PLAN | **PM13** |
| #Hs09 | Chemical upset condition ruptures vessel, pipeline, or wellhead, releasing CO2 at IDLH rate. | NONCONTAINMENT RESPONSE PLAN | **PM13** |
| #Ci23 | Increased national urgency to execute CCS quickly forces injection to be done with less baseline data or less monitoring than planned. | GOAL TRIAGE | **PM14** |
| #Ci24 | Increased national urgency to execute CCS cheaply forces injection to be done with less baseline data or less monitoring than planned. | GOAL TRIAGE | **PM14** |
| #Fn01 | Price escalation for materials or consumables requires enough budget that not all project goals can be met. | GOAL TRIAGE | **PM14** |
| #Fn02 | Price escalation for labor, services, lease/royalty payments, or insurance requires enough budget that not all goals can be met. | GOAL TRIAGE | **PM14** |
| #Rs02 | Overall research budget is reduced regardless of cause, thus not all goals can be met. | GOAL TRIAGE | **PM14** |
| #Ci01 | Public concern about water mobilizes opposition that impedes project activities, arising without a specific triggering event. | COMMUNICATIONS PLAN | **PM15** |
| #Ci06 | An unexpected situation arises that is not technically impacting, but the appearance of unpreparedness spurs public concern ... | COMMUNICATIONS PLAN | **PM15** |
| #Ci10 | Public fears of CO2 injection or monitoring, though technically not well founded, create opposition that impedes operations. | COMMUNICATIONS PLAN | **PM15** |
| #Ci16 | Model results that become public suggest plume leaving site. Adjacent land owners decide to test laws of pore space ownership resulting in lengthy court cases. Project on hold for lengthy period of time. | COMMUNICATIONS PLAN | **PM15** |
| #Fn09 | NGOs take a united public stand against carbon sequestration, causing delays leading to price escalation. | COMMUNICATIONS PLAN | **PM15** |
| #Ci12 | Small earthquakes happen and though they are scientifically shown to be unrelated to project, public/media concern forces project to devote time to debunking rumors. | SITE SELECTION AND AOR | **SR01** |
| #Mg01 | CO2 migrates to undiscovered unsealed orphan well that provides a leakage path. | SITE SELECTION AND AOR | **SR01** |
| #Op02 | Severe weather closes roads causing delay in operations resulting in lost rig days. | SITE SELECTION AND AOR | **SR01** |
| #Rs09 | Public opposition based on feared impact to land use (e.g. farming) impedes using monitoring method. | SITE SELECTION AND AOR | **SR01** |
| #Ci05 | Non-project-related groundwater change is ascribed to project, spurring public concern ... | USDW BASELINE CHEMISTRY | **SR02** |

| Scen ID# | SCENARIO | Assigned **Risk Response Action Group** | RRAG id |
|---|---|---|---|
| #Hs04 | CO2 acidifies groundwater which mobilizes metals already present within the USDW lithology. | USDW BASELINE CHEMISTRY | **SR02** |
| #Hs15 | Groundwater with high CO2, metals, toxic gas, or acid content moves into shallow aquifer or surface water body. | MVA PLAN | **SR06** |
| #Rs04 | Subsurface condition impairs seismic imaging, groundwater sampling, logging, or other data acquisition, impairing ability to track plume. | MVA PLAN | **SR06** |
| #Rs05 | Subsurface condition controls plume migration into area where monitoring ability is impaired. | MVA PLAN | **SR06** |
| #Rs08 | Land use (e.g. planted fields) physically impedes using monitoring method. | MVA PLAN | **SR06** |
| #Rs13 | Weather conditions prevent using a monitoring method or retrieving data at a critical point in time. | MVA PLAN | **SR06** |
| #Ci11 | Project surface operations (e.g. seismic data acquisition) impair other land uses, or cause physical damage, such that public opposition arises. | SURFACE SEISMIC | **SR08** |
| #Op10 | During the annual seismic survey the vibrations cause an aging water or gas line to fail resulting in a water or gas leak and an outage in the community. This could be associated as well with the danger of a gas explosion from the gas leak. | SURFACE SEISMIC | **SR08** |
| #Hs03 | CO2-acidified brine moves via wellbore or fault, dissolving metals along the way, into USDW. | STATIC EARTH MODEL | **SR09** |
| #Mg05 | CO2 exits storage reservoir and crosses caprock via fault or fracture. | STATIC EARTH MODEL | **SR09** |
| #Mg06 | Subsurface condition causes plume migration beneath sensitive area or causes unexpectedly long distance migration. | STATIC EARTH MODEL | **SR09** |
| #Fn06 | Plume travels off property controlled by project partner, incurring legal action or payment demand by adjacent property owner. | SIMULATION PLAN | **SR10** |
| #Hs01 | CO2 laterally displaces Mt. Simon brine with dissolved metals, moving brine to where Mt. Simon had freshwater / is a USDW. | SIMULATION PLAN | **SR10** |
| #Rs12 | Incorrect or inadequate physical process models incorrectly predict plume fate, causing mistargeted data acquisition. | SIMULATION PLAN | **SR10** |
| #Sf04 | Drilling or mining for unrelated purpose encounters unexpected high pressure fluid, damaging equipment and/or resource. | SIMULATION PLAN | **SR10** |

# APPENDIX C: RISK TREATMENTS (RISK RESPONSE ACTIONS) AND DOCUMENTATION SUMMARY

| RRAG | RRA sqno | Risk Treatment / Risk Response Action (RRA) | Due By *date or event* | Done Date |
|---|---|---|---|---|
| OE01 | 00001 | Prepare a detailed QHSE Site Plan to cover construction, downhole, drilling/completion, and seismic activities not involving CO2 that includes at least the following: | 2009 | 2009 |
| OE01 | 00002 | Define minimum initial and recurrent training requirements for workers and visitors, confined-space activities, weather hazards, motor vehicle safety, on-site and off-site work, and all other QHSE issues that might arise pertinent to these activities. | pre Injection | 2008 |
| OE01 | 00003 | Ensure that no one works at site before completing the provided safety training. | pre Injection | 2008 |
| OE01 | 00007 | Ensure that Journey Management and Site Traffic Plan are written, communicated, and trained for, and that performance per JMP and STP is tracked. | pre Injection | 2008 |
| OE01 | 00008 | Ensure that performance per JMP and STP is tracked. | pre Injection | 2008 |
| OE01 | 00010 | Ensure that plan includes a Bridging Document that covers the specific organizations involved in these activities. | 2009 | 2008 |
| OE02 | 00004 | During project phases and near locations where high-pressure CO2 is present, identify hazards specific to its presence, and incorporate safety procedures into Surface QHSE Site plan. | pre Injection | risk OK'd per PI, Nov 2011 |
| OE02 | 00006 | Ensure that procedures, training, and available equipment support the prevention of anyone working in a confined space without proper ventilation and active oversight. | pre Injection | 2-May-11 |
| OE02 | 00011 | Prepare detailed Surface QHSE Site Plan to cover surface and injection operations. | pre Injection | risk OK'd per PI, Nov 2011 |
| OE02 | 00012 | Ensure that Surface QHSE Site Plan covers minimum initial and recurrent training requirements for workers and visitors, weather hazards, motor vehicle safety, on-site and off-site work, all areas of site, site office, and injection-well end of pipeline, and safety equipment to be stored at site office. | pre Injection | risk OK'd per PI, Nov 2011 |
| OE02 | 00013 | Perform risk-reduction measures specific to scenarios of high-flow CO2 release, especially in proximity to confined spaces including the injection-well cellar and semi-confined spaces such as berm-bounded areas. | pre Injection | risk OK'd per PI, Nov 2011 |
| OE02 | 00014 | Ensure that no one works at site before completing the provided safety training. | pre Injection | risk OK'd per PI, Nov 2011 |
| OE02 | 00015 | Ensure the presence and continuous functionality of an alarm system to warn persons on and near site in the event of significant above-background CO2 in air, with planned project-specific alert levels and response protocols, and with monitoring locations to include EC tower, well pad, and site office. | pre Injection | risk OK'd per PI, Nov 2011 |
| OE02 | 00017 | Ensure that Journey Management and Site Traffic Plan are written, communicated, and trained for, and that performance per JMP and STP is tracked. | pre Injection | risk OK'd per PI, Nov 2011 |
| OE02 | 00018 | Prepare a crisis call list ***and review it monthly for currency.*** | pre Injection | risk OK'd per PI, Nov 2011 |

| RRAG | RRA sqno | Risk Treatment / Risk Response Action (RRA) | Due By *date or event* | Done Date |
|---|---|---|---|---|
| OE02 | 00019 | Ensure that plan includes a Bridging Document that covers the specific organizations involved in these activities. | pre Injection | risk OK'd per PI, Nov 2011 |
| OE02 | 00020 | Include HAZOP exercises as necessary. | pre Injection | 17-Sep-09 |
| OE02 | 01008 | Review Subsurface QHSE Plan, and ensure that any project QHSE aspect that is not specifically addressed by the Subsurface QHSE Plan is addressed by the Surface QHSE Site Plan. | pre Injection | risk OK'd per PI, Nov 2011 |
| OE03 | 00021 | Develop site security plan that includes active 24/7 security, fencing, signage, warning devices, sensors, alarms, physical barriers to prevent rupture from accidental impact (e.g. by truck) upon key components including wellhead, pipeline. (Note: removed "EC tower" from list). | pre Injection | risk OK'd per PI, Nov 2011 |
| OE04 | 00022 | Ensure that a detailed drilling plan is in place covering at least these elements: | 2009 | 2009 |
| OE04 | 00023 | Construction plan (suited as appropriate to conditions pre-injection, during injection, or after well control operations); | pre each well | 31-Jul-09 |
| OE04 | 00024 | Drillsite security; | pre each well | 2008 |
| OE04 | 00025 | Wellbore security; | pre-moveoff per well | recurrent / ongoing |
| OE04 | 00026 | Operations to reestablish normal mud balance and/or circulating conditions (regaining control within several hours at most); | pre each well | 21-Aug-10 |
| OE04 | 00028 | Equipment and personnel training requirements; | pre each well | 25-Apr-11 |
| OE04 | 00029 | Drill Well On Paper and Complete Well On Paper (DWOP and CWOP) exercises. | pre each well | 2008 |
| OE05 | 00030 | Write a detailed plan for completing any injection or monitoring well, covering at least these elements: | not set, timely done | 23-Sep-09 |
| OE05 | 00031 | Inspection and verification of wellbore mechanical integrity, and reconfirmation as necessary if significant time has passed before starting injection; | not set, timely done | 1-Oct-09 |
| OE05 | 00032 | regulatory required specifications; additional logging needs; | not set, timely done | 1-May-10 |
| OE05 | 00033 | timetable for ordering required equipment; | not set, timely done | 20-Sep-10 |
| OE06 | 00034 | Write a detailed plan for completing any injection or monitoring well, covering at least these elements: | not set, timely done | 1-Nov-09 |
| OE06 | 00035 | Inspection and verification of wellbore mechanical integrity, and reconfirmation as necessary if significant time has passed before starting injection; | not set, timely done | 3-Feb-10 |
| OE06 | 00036 | regulatory required specifications; | not set, timely done | Aug-09 |
| OE06 | 00037 | additional logging needs; | not set, timely done | Aug-09 |
| OE06 | 00038 | timetable for ordering required equipment; | 30-Sep-11 | 30-Sep-11 |

| RRAG | RRA sqno | Risk Treatment / Risk Response Action (RRA) | Due By *date or event* | Done Date |
|---|---|---|---|---|
| OE07 | 00039 | Write a detailed plan for completing any monitoring well, covering at least these elements: | pre each well | 17-Sep-10 |
| OE07 | 00040 | Inspection and verification of wellbore mechanical integrity | pre each well | 15-Jun-10 |
| OE07 | 00041 | regulatory required specifications; | pre each well | 8-Apr-10 |
| OE07 | 00042 | additional logging needs; | pre each well | 15-Jun-10 |
| OE07 | 00043 | timetable for ordering required equipment; | not set, timely done | 15-May-10 |
| OE08 | 00044 | To counter a potentially corrosive downhole environment that may feature CO2-saturated brine, | not set, timely done | 28-Nov-09 |
| OE08 | 00045 | use chrome steel casing and special CO2-resistant cement, | not set, timely done | 4-May-10 |
| OE08 | 00046 | and ensure that cement placement and Mechanical Integrity Testing (MIT) are performed to the highest standards. | not set, timely done | 4-May-10 |
| OE08 | 00047 | Ensure that CO2-resistant well construction characteristics are accurately and thoroughly represented in the final submitted Completion Report. | not set, timely done | 1-May-10 |
| OE09 | 00058 | Envelope Maximization: To minimize risk from negatively impacting excursions from optimal operating conditions, design all mechanical systems to operate under as wide a range of conditions as may be practically achievable. Further details are specified: | pre Injection | 25-Oct-10 |
| OE09 | 00059 | Include dehydration, purification, compression, heating/cooling, pipeline, tanks, and distribution systems, | pre Injection | 25-Oct-10 |
| OE09 | 00060 | Address CO2 delivery at the wellhead at a range of pressures, temperatures, purities, and flow rates, up to the maximum range of each individual parameter, | pre Injection | 25-Oct-10 |
| OE09 | 00061 | Address maximum ranges as allowed by various limitations including downhole injection pressure, maximum/minimum allowed temperature profile in the well, and corrosion/geochemistry/HSE/legal considerations on the injectate composition; | pre Injection | 25-Oct-10 |
| OE09 | 00062 | Avoid design elements that would operate poorly outside of the optimal design point, | pre Injection | 25-Oct-10 |
| OE09 | 00063 | Characterize the impact on capital and operating expenditures of operational variations within the proposed operating envelope, | pre Injection | 20-Dec-10 |
| OE09 | 00064 | Provide a carbon and energy balance of the injection system operating at design conditions, | pre Injection | 20-Dec-10 |
| OE09 | 00065 | Ensure that the conditions of delivery of CO2 from the separation/transport equipment and the supply of CO2 to the wells are consistent with the respective components' design specifications, | pre Injection | 3-Jun-10 |
| OE09 | 00066 | Refine design specifications early, ensuring that newly acquired information on the geological environment or the wells' completions is quickly reviewed for potential effect upon delivery-system design, and communicated as needed. | pre Injection | 6-May-10 |

| RRAG | RRA sqno | Risk Treatment / Risk Response Action (RRA) | Due By *date or event* | Done Date |
|---|---|---|---|---|
| OE09 | 00067 | Envelope Definition: To support ongoing injection operation within the range of expected “safe” conditions, create a concise statement (supported by clear graphics and charts) of the estimated "operating range envelope", containing at least these elements and characteristics: | pre Injection | 25-Oct-10 |
| OE09 | 00068 | Foundation in calculations and simulations based on best current physico-chemical understandings, | pre Injection | 20-Dec-10 |
| OE09 | 00069 | Description of the boundaries to the ranges of conditions within which the system can be allowed to vary without significant infrastructure implications and/or without significant subsurface implications. | pre Injection | 25-Oct-10 |
| OE09 | 00070 | Description of the likely implications of operations outside the described safe envelope. | pre Injection | 31-May-11 |
| OE09 | 00071 | Justification (from modeling or passive/active control mechanisms) that the delivery/transport/subsurface systems can operate in a stable and safe manner together across the whole operating range envelope, | pre Injection | 31-May-11 |
| OE09 | 00072 | Demonstration that that the system can be controlled or shut-down safely in case of departure of any sub-system from the operating range envelope, | pre Injection | risk OK'd per PI, Nov 2011 |
| OE09 | 00073 | Demonstration that proposed process monitoring and control equipment are sufficient to define the system state within the operating range envelope, and that unwanted departures can be safely identified and corrected. | pre Injection | risk OK'd per PI, Nov 2011 |
| OE10 | 00048 | Develop a site-specific injection operating plan that includes at least these elements: | pre Injection | risk OK'd per PI, Nov 2011 |
| OE10 | 00049 | Plan technical basis informed by results from Integrated Asset Modeling, | pre Injection | 20-Dec-10 |
| OE10 | 00050 | Constant injection operations, planned shutins, and unplanned shutins, | pre Injection | risk OK'd per PI, Nov 2011 |
| OE10 | 00051 | Protecting operating equipment and adjusting operations to account for weather stresses (e.g. output temperature) and expected inflow variations, | pre Injection | 6-May-10 |
| OE10 | 00052 | Plan, procedures, and equipment availability for injection turnaround within one day, in any normally expected weather conditions. | pre Injection | 31-May-11 |
| OE10 | 00053 | Provisions for reverse well flow of CO2 and brine, reverse pipeline flow and stack venting of CO2, and brine separation and disposal. | pre Injection | risk OK'd per PI, Nov 2011 |
| OE10 | 00054 | Plan, procedures, and equipment availability for immediately shutting in the injection well and suspending CO2 injection for an indefinite period. | pre Injection | risk OK'd per PI, Nov 2011 |
| OE10 | 00055 | Presence and continuous functionality of a shutoff mechanism on the compression facility, that responds automatically (plus manually) to significant changes in pressure or flow rate; and that protects compressors from sudden loss of pipeline pressure. | pre Injection | 29-Jan-11 |
| OE10 | 00056 | Procedures and equipment availability to enable stopping injection on short notice. | pre Injection | 29-Jan-11 |
| OE10 | 00057 | Preliminary plan for a Startup-Shutin On Paper (SUSIOP) exercise comparable to existing DWOP and CWOP exercises. | pre Injection | risk OK'd per PI, Nov 2011 |
| OE11 | 00074 | For each project mechanical system (including all project equipment assets), develop a list of critical equipment, maintenance schedules, and equipment-acquisition availability and replacement lead times (procurement plus installation). | 30-Jun-12 | |
| OE11 | 00075 | Ensure that both operational and monitoring equipment are included in equipment sparing plan. | 30-Jun-12 | |

| RRAG | RRA sqno | Risk Treatment / Risk Response Action (RRA) | Due By *date or event* | Done Date |
|---|---|---|---|---|
| OE11 | 00076 | Estimate the loss severity / hazard / risk associated with loss/malfunction of each item of critical equipment, and establish a risk-based, item-specific sparing policy. | pre Injection | risk OK'd per PI, Nov 2011 |
| OE11 | 00077 | Identify suppliers of backup equipment. | pre Injection | 20-Dec-10 |
| OE12 | 00078 | For the purpose of enabling quick response to protect HSE in the event of a hazardous condition, develop a plan to continuously monitor live data streams from sensors measuring operational and key surface environmental parameters. Ensure the plan addresses at least these elements: | pre Injection | risk OK'd per PI, Nov 2011 |
| OE12 | 00079 | Automated vs. manned monitoring, | pre Injection | risk OK'd per PI, Nov 2011 |
| OE12 | 00080 | Visual inspections, | pre Injection | risk OK'd per PI, Nov 2011 |
| OE12 | 00081 | Implications of anticipated false-positive and false-negative rates, | pre Injection | risk OK'd per PI, Nov 2011 |
| OE12 | 00082 | Predefinition of sensor readings that trigger shutdown of any part of injection system, | pre Injection | 31-May-11 |
| OE12 | 00083 | Information flow and communications issues, | pre Injection | risk OK'd per PI, Nov 2011 |
| OE12 | 00084 | Both physical and stakeholder (human relations and communications) impacts of notification and shutdown scenarios, | pre Injection | risk OK'd per PI, Nov 2011 |
| OE12 | 00085 | Lessons Learned from a SUSIOP exercise, if carried out. | pre Injection | 20-Dec-10 |
| OE12 | 01006 | Maintenance of system safety (preventing hazardous conditions and damaging impacts) in case of malfunction of electronic or computer-controlled systems, | pre Injection | 1-Feb-11 |
| PM01 | 00086 | Develop and execute risk management process that includes at least these elements: | pre Injection | Dec-07 |
| PM01 | 00087 | Risk identification, | pre Injection | 20-Feb-08 |
| PM01 | 00088 | Risk evaluation, | pre Injection | Jun-08 |
| PM01 | 00089 | Actions to minimize risks; | pre Injection | 29-Jun-10 |
| PM01 | 00090 | Documentation and communications that provide assurance to stakeholders, | not set, timely done | Sep-10 |
| PM01 | 00091 | Documentation adequate to demonstrate that best practices (engineering and geological) have been followed. (note rewording from prior - was "legal proceedings"). | Jan-11 | Jul-11 |
| PM01 | 00092 | An IT risk-management solution, e.g. similar to or an adaptation of the Schlumberger Quest system. | Dec-10 | Jul-11 |
| PM01 | 00240 | Consider moving facilities and injection-related tasks in OE01 and OE04 to other RRAGs for which "R" person is on site (ISGS, Carb Svcs IL team, ADM). | pre Injection | Dec-10 |
| PM01 | 00241 | Identify which RRAs are critical to complete before injection start. | 15-Dec-10 | 16-Sep-10 |
| PM01 | 00242 | Documentation and communications that provide assurance to stakeholders about risk evaluation and management | Jan-11 | risk OK'd per PI, Nov 2011 |

| RRAG | RRA sqno | Risk Treatment / Risk Response Action (RRA) | Due By *date or event* | Done Date |
|---|---|---|---|---|
| PM02 | 00093 | Write permitting and compliance plan. Details are specified. | pre Injection | risk OK'd per PI, Nov 2011 |
| PM02 | 00094 | Complete listing of needed permits and potential permit modifications; | 30-Jun-12 | |
| PM02 | 00095 | Listing of permits potentially needed at short notice (such as for workovers) and their information requirements; | 30-Jun-12 | |
| PM02 | 00096 | Permit-issuance schedule/forecast; | 30-Jun-12 | |
| PM02 | 00097 | Resource estimate for permitting-agency communications and support; | 30-Jun-12 | |
| PM02 | 00098 | Tables and schedules documenting all permit requirements and their assignment for execution (RACI); | pre Injection | risk OK'd per PI, Nov 2011 |
| PM02 | 00099 | Archiving/recordkeeping plan for agency communications regarding permits; | pre Injection | risk OK'd per PI, Nov 2011 |
| PM02 | 00100 | Development of completion report plan and assignment for execution (RACI). | 30-Jun-12 | |
| PM03 | 00101 | Establish written protocols for management of all types of data collected by project, including continuous/episodic/irregular; numerical/textual/image/audio; originating in loggers/laptops/mainframes/cameras/handwritten; etc. | pre Injection | 18-Oct-11 |
| PM03 | 00102 | Include procedures to prevent premature distribution of raw and/or unreviewed data. | pre Injection | 15-Apr-11 |
| PM03 | 00103 | Ensure plan includes guidelines and policy for data collection and upload; technical (QA/QC) and administrative review; storage, archiving, general and limited distribution, and MGSC partner access. | pre Injection | 18-Oct-11 |
| PM03 | 00104 | Ensure data is kept accessible to project partners as needed and as consistent with policy. | pre Injection | 18-Oct-11 |
| PM03 | 00105 | Establish criteria for publishing and public data release, consistent with Communications Plan. | pre Injection | 15-Apr-11 |
| PM03 | 00106 | Consider adapting existing IT and IPM solutions such Schlumberger "InterAct". | pre Injection | 15-Apr-11 |
| PM03 | 00107 | Maintain data in the simplest common format in which data integrity can be preserved, e.g., character delimited format, platform independent, on robust storage media. | pre Injection | risk OK'd per PI, Nov 2011 |
| PM03 | 01001 | Maintain records of project activities to support defense against not-unlikely claims of damage, in conjunction with baseline environmental measurements. | pre Injection | risk OK'd per PI, Nov 2011 |
| PM04 | 00108 | Ensure that budgeting plan addresses at least these elements: | 30-Jun-12 | |
| PM04 | 00109 | Ongoing awareness of funding status with funding agency; | 30-Jun-12 | |
| PM04 | 00110 | Ongoing efforts to reduce cost- and budget-sourced risks; | 30-Jun-12 | |
| PM04 | 00111 | Mission Sensitivity assessment of each budget area to budget changes; | 30-Jun-12 | |
| PM04 | 00112 | Cost estimates (to partnership and/or partners) for potential events with significant cost implications such as groundwater contamination; | not set, timely done | Jul-11 |
| PM04 | 00113 | Review of possible insurance coverages and costs; | 30-Jun-12 | |

| RRAG | RRA sqno | Risk Treatment / Risk Response Action (RRA) | Due By *date or event* | Done Date |
|---|---|---|---|---|
| PM04 | 00114 | Cost-control strategies in subcontractor relationships; | 30-Jun-12 | |
| PM04 | 00115 | Quarterly and event-driven AFE review; | 30-Jun-12 | |
| PM04 | 00116 | Review of potential cost-share arrangements with other RCSP partnerships; | 30-Jun-12 | |
| PM04 | 00117 | Cost-based review of Sparing and Procurement plans; | 30-Jun-12 | |
| PM04 | 00118 | Contingency planning for unusual monitoring, emergency repairs, or other purposes; | pre Injection | risk OK'd per PI, Nov 2011 |
| PM04 | 00119 | Contingency planning for general budget shortfalls. | 30-Jun-12 | |
| PM05 | 00120 | Ensure that Project Management Plan addresses at least these elements: | not set, timely done | 18-Mar-08 |
| PM05 | 00121 | Project overview, organization, and structure; | not set, timely done | 18-Mar-08 |
| PM05 | 00122 | Risk management approach; | pre Injection | 18-Mar-08 |
| PM05 | 00123 | Project schedule, milestones, and deliverables; | not set, timely done | 18-Mar-08 |
| PM05 | 00124 | Timing flexibility / criticality; | not set, timely done | 18-Mar-08 |
| PM05 | 00125 | Budget summary and cost plan; | not set, timely done | 18-Mar-08 |
| PM05 | 00126 | Success criteria and decision points; | not set, timely done | 18-Mar-08 |
| PM05 | 00127 | Regular staff communications; | 30-Jun-12 | |
| PM05 | 00128 | Regular coordination among technical and task leaders; | pre Injection | risk OK'd per PI, Nov 2011 |
| PM05 | 00129 | Staff continuity / expertise maintenance; | not set, timely done | 18-Mar-08 |
| PM05 | 00130 | Verification of critical-task completion. | pre Injection | 18-Mar-08 |
| PM05 | 00131 | Organize the formation of self-run, self-led technical teams to review data and plans and to make timely technical recommendations, for certain project activities for which concentrated and diverse technical expertise is needed to support decisionmaking – especially where there may be competing objectives. | 30-Jun-12 | |
| PM05 | 00132 | Ensure that teams are organized around at least the following activities: seismic survey design, core-point definition and core disposition, unanticipated events, and injectivity backup and goal triage (see separate RRAG's). | 30-Jun-12 | |
| PM08 | 00142 | Ensure that a project communications policy addresses at least these elements: | not set, timely done | 15-Jan-09 |
| PM08 | 00143 | Release of raw data and modeling results, | not set, timely done | 15-Jan-09 |

| RRAG | RRA sqno | Risk Treatment / Risk Response Action (RRA) | Due By *date or event* | Done Date |
|---|---|---|---|---|
| PM08 | 00144 | Communications about possible negative events, | not set, timely done | 15-Jan-09 |
| PM08 | 00145 | Linkage of project with particular societal goals and/or industries, | not set, timely done | 15-Jan-09 |
| PM08 | 00146 | Internal publishing and execution of the communications policy itself, | not set, timely done | 15-Jan-09 |
| PM08 | 00147 | Staff training, | not set, timely done | 15-Jan-09 |
| PM08 | 00148 | Review processes. | not set, timely done | 15-Jan-09 |
| PM12 | 00178 | If characterization and/or operations data appear to show lower expected or actual injectivity than needed, organize project group to evaluate options including data reacquisition / verification, additional injection points (in same or different well), controlled fracturing, proceeding with lower than desired injectivity, and others. | pre Injection | 13-Nov-10 |
| PM13 | 00179 | Write contingency plans for unexpected $CO_2$ movement that address at least the following scenarios (x): | pre Injection | risk OK'd per PI, Nov 2011 |
| PM13 | 00180 | (x) Data showing CO2 laterally migrated significantly beyond expectations based on models; and | pre Injection | 25-Aug-11 |
| PM13 | 00181 | (x) CO2 present in non-design locations in injection-well casing or annulus. | pre Injection | 25-Aug-11 |
| PM13 | 00182 | Address public communications for these events in this plan or the general Communications Plan. | pre Injection | risk OK'd per PI, Nov 2011 |
| PM13 | 01002 | (x) Significant atmospheric CO2 release ~~requiring dispersal;~~ | pre Injection | risk OK'd per PI, Nov 2011 |
| PM13 | 01003 | (x) Discovery of a monitoring well or non-project well providing a leakage pathway for CO2 or brine; | pre Injection | 25-Aug-11 |
| PM13 | 01004 | (x) Data showing CO2 above the Eau Claire caprock; | pre Injection | 25-Aug-11 |
| PM13 | 01005 | (x) Pipeline break, emphasizing protection of injection wellhead and completion. | pre Injection | 29-Jan-11 |
| PM13 | 01007 | (x) Abrupt wellhead rupture/shear below master valve, caused by accident or intention; | pre Injection | risk OK'd per PI, Nov 2011 |
| PM14 | 00183 | Ensure that specific individuals are tasked to monitor progress toward research goals, and to communicate to project managers should goal attainment be in doubt. | 30-Jun-12 | |
| PM14 | 00184 | Develop plan for convening project leaders, partner representatives, and NETL representatives to prioritize goals and potential responses, in the event that goal attainment appears significantly threatened. | 30-Jun-12 | |
| PM15 | 00149 | Develop a Communications (Comms) Plan to establish preparedness for potential comms challenges, and to timely and accurately convey project info to target audiences. | pre Injection | 15-Jan-09 |
| PM15 | 00150 | Include Comms Plan element: Text statement of overall purpose and strategy of the Comms Plan. | not set, timely done | 15-Jan-09 |
| PM15 | 00151 | Include Comms Plan element: Descriptions of MGSC consortium, IBDP project, and target audiences. | not set, timely done | 15-Jan-09 |

| RRAG | RRA sqno | Risk Treatment / Risk Response Action (RRA) | Due By *date or event* | Done Date |
|---|---|---|---|---|
| PM15 | 00152 | Include Comms Plan element: Guidance text for Listening, Information Gathering, and Influencing; Site Visits; Publications; Crisis Communications. | not set, timely done | 15-Jan-09 |
| PM15 | 00153 | Include Comms Plan element: Prepare policy documents addressing general safety, data distribution, photography, and publications review process. | not set, timely done | 15-Jan-09 |
| PM15 | 00154 | Include Comms Plan element: Prepare and maintain documents that list project Talking Points and Answers to Frequently Asked Questions. | pre Injection | 15-Jan-09 |
| PM15 | 00155 | Include Comms Plan element: Plan for frequent monitoring of and content supply to mainstream media and social electronic information media. | 30-Jun-12 | |
| PM15 | 00156 | Include Comms Plan element: Maintain a current, publishable information set on project activities, project status, characterization data, and monitoring data. | not set, timely done | 15-Jan-09 |
| PM15 | 00157 | Include Comms Plan element: Create a publications record that shows project progress and progressively improved subsurface and plume knowledge. | not set, timely done | 30-Jun-09 |
| PM15 | 00158 | Include Comms Plan element: Documentation of and communications about project awareness of, development of, and adherence to industry best practices. | not set, timely done | 1-Dec-10 |
| PM15 | 00159 | Include Comms Plan element: Plan for timely and thorough responses to challenges (in permitting or other contexts). | pre Injection | 15-Jan-09 |
| PM15 | 00160 | Include Comms Plan element: Plan for project-partner participation in project meetings, conferences, comms-team briefings, DOE OWG, and industry-group meetings. | not set, timely done | 15-Jan-09 |
| PM15 | 00161 | Include Comms Plan element: Description of groundwater usage within AOR; strategy and plans for relations with groundwater stakeholder organizations; publication of project's groundwater protection efforts. | not set, timely done | 15-Jan-09 |
| SR01 | 00185 | Gather regional information pertinent to site selection and review, including many of the following depending on associated risks: | not set, timely done | done pre-spud INJ1 |
| SR01 | 00186 | Hydrochemically and geochemically characterize USDWs; | not set, timely done | done pre-spud INJ1 |
| SR01 | 00187 | Ensure that lithostratigraphic data support that effective primary and secondary caprocks extend well beyond AOR; | not set, timely done | done pre-spud INJ1 |
| SR01 | 00188 | Identify and ensure that permitting records and anecdotal drilling history support negligible risk of unknown abandoned wellbores; | not set, timely done | done pre-spud INJ1 |
| SR01 | 00189 | Physically inspect all known abandoned wellbores (or their sites/locations) that reach the aquifer overlying the primary caprock; | not set, timely done | done pre-spud INJ1 |
| SR01 | 00190 | Chemically characterize geologic media through which CO2-induced movement may occur; | not set, timely done | done pre-spud INJ1 |
| SR01 | 00191 | Characterize historical environmental conditions that might influence the interpretation of monitoring data on the site; | not set, timely done | done pre-spud INJ1 |
| SR01 | 00192 | Establish a well sampled baseline of environmental monitoring data (and continue to monitor baseline parameters in case of project delay); | not set, timely done | done pre-spud INJ1 |
| SR01 | 00193 | Review groundwater records for potential changes induced near natural-gas storage facilities. | not set, timely done | done pre-spud INJ1 |

| RRAG | RRA sqno | Risk Treatment / Risk Response Action (RRA) | Due By *date or event* | Done Date |
|---|---|---|---|---|
| SR02 | 00194 | Establish baseline geochemical and hydrochemical attributes and flow-related characteristics of USDW's within AOR. | pre Injection | 17-Dec-10 |
| SR02 | 00195 | Ensure USDW characterization is adequate to support a later demonstration (if justified by data) that groundwater resources have not been negatively impacted by project. | pre Injection | 17-Dec-10 |
| SR02 | 00196 | Establish baseline (including variation ranges) pH, and concentrations of constituents as specified in applicable permit conditions or of potential concern. | pre Injection | 17-Dec-10 |
| SR03 | 00197 | Write a detailed plan for data acquisition in wells that penetrate or approach the target injection zone, that delivers required information to the Well Bid Package and Well AFE, and addresses at least the following elements: | Sep-10 | done pre-spud INJ1 |
| SR03 | 00198 | Satisfaction of risk management, regulatory, operational, and research goals; | Sep-10 | done pre-spud INJ1 |
| SR03 | 00199 | Data-acquisition implications for well design and instrumentation; | not set, timely done | Jun-10 |
| SR03 | 00200 | Open-hole and cased-hole logging; | Sep-10 | done pre-spud INJ1 |
| SR03 | 00201 | Cuttings samples, and whole and sidewall cores; | Sep-10 | done pre-spud INJ1 |
| SR03 | 00202 | Fluid sampling, including drillstem testing; | Sep-10 | done pre-spud INJ1 |
| SR03 | 00203 | Coordination of objectives with the overall MVA plan; and | Sep-10 | done pre-spud INJ1 |
| SR03 | 00204 | Data delivery to the Data Management System. | Sep-10 | done pre-spud INJ1 |
| SR06 | 00027 | Observations/data collection plan during a potential leakage event; | pre Injection | risk OK'd per PI, Nov 2011 |
| SR06 | 00222 | Write an MVA plan that covers all methods and instrumentation used to monitor effects of CO2 injection and plume migration; | pre Injection | risk OK'd per PI, Nov 2011 |
| SR06 | 00223 | Address pre-deployment testing, deployment procedures, procedures and frequencies for data collection and sampling, data delivery to Data Management System, and sparing policy for long-lead-time instruments | pre Injection | risk OK'd per PI, Nov 2011 |
| SR06 | 00224 | Review planned MVA techniques to ensure they will provide early detection of near-wellbore CO2 movement out of the reservoir or vertically up the borehole. | pre Injection | risk OK'd per PI, Nov 2011 |
| SR06 | 00225 | Determine if monitoring data gaps exist that affect the ability to meet data needs for the purposes of HSE protection, plume simulation, and research; | pre Injection | risk OK'd per PI, Nov 2011 |
| SR06 | 00226 | After deployment of field instruments, verify data collection integrity soon and frequently. | pre Injection | risk OK'd per PI, Nov 2011 |
| SR06 | 00227 | For data automatically transmitted from remote locations, only expand length of time between data transmissions (or field visits) once data collection and transmission integrity have been demonstrated. | pre Injection | risk OK'd per PI, Nov 2011 |
| SR06 | 00228 | Periodically review data and plume-simulation forecasts to enhance/adjust monitoring strategy. | pre Injection | risk OK'd per PI, Nov 2011 |
| SR06 | 00229 | If observations or simulations indicate plume migration into area where monitoring is or will be impaired, formulate revised operational and/or surface-access plans. | pre Injection | risk OK'd per PI, Nov 2011 |

| RRAG | RRA sqno | Risk Treatment / Risk Response Action (RRA) | Due By *date or event* | Done Date |
|---|---|---|---|---|
| SR06 | 00230 | Ensure that elements of the MVA program (e.g. chemical sampling and micro- and passive seismic monitoring) and simulation program are adequate to demonstrate deployment of best practices (engineering / geological / geophysical). | pre Injection | risk OK'd per PI, Nov 2011 |
| SR06 | 00231 | As part of overall MVA plan, assess impact of seasonal, weather, and land-use variability (e.g., cropping, development) on data collection efforts, monitoring equipment, and operational infrastructure. | pre Injection | risk OK'd per PI, Nov 2011 |
| SR06 | 00232 | Ensure high data quality by selecting and verifying monitoring equipment, methods, and schedule. | pre Injection | risk OK'd per PI, Nov 2011 |
| SR08 | 00005 | Confirm that seismic operator (Geco) has developed and adheres to an adequate QHSE plan. | pre Injection | risk OK'd per PI, Nov 2011 |
| SR08 | 00233 | Conduct repeated 2D and/or 3D seismic monitoring to track the CO2 plume, provide information for simulations/modeling, and identify faults and fracture zones. | 30-Jun-12 | recurrent / ongoing |
| SR08 | 00234 | Before each field acquisition, conduct seismic plan on paper (SPOP) exercise to anticipate problems and QHSE issues and to optimize data quality. | not set, timely done | 31-Dec-09 |
| SR09 | 00235 | Develop plan for constructing and refining a computer-hosted static earth model including at least these elements: | 15-Mar-11 | 1-Apr-11 |
| SR09 | 00236 | model selection and design, data sources, specifications for review and incorporation of new measured data, magnitude and significance of geological uncertainties in terms of plume-migration influence and project risk, and coordination with the Data Management System for geologic characterization data. | 30-Jun-12 | recurrent / ongoing |
| SR10 | 00237 | Develop plan for performing plume simulations and comparing their results. Further detail is specified. | Sep-10 | replan after 3D 2012 |
| SR10 | 00238 | Include specifications for review and incorporation of measured data, model selection and design, nature of output, sensitivity analysis, documentation, and coordination with the Data Management System for geologic characterization data. | Sep-10 | replan after 3D 2012 |
| SR10 | 00239 | Ensure documentation addresses the range of possible results, the level of confidence in the predicted plume behavior, the likelihood of unexpected behavior, and the criticality of potential additional measured data from existing or new monitoring locations, and includes information to evaluate and compare different simulations. | Sep-10 | replan after 3D 2012 |

*In addition to due dates and done dates as shown, data items tracked per RRA include RACI personal assignments (see Appendix C), text descriptions of actual actions completed, variances, and links to documentation files including simulation records and completed workplans.*

# APPENDIX D: "RACI" CHART

| RRAG sqno | RRAG FID | Risk Response Action Group (RRAG) | R Responsible | A Accountable | C Communicate With | I Inform |
|---|---|---|---|---|---|---|
| 1 | OE01 | QHSE PLAN: Subsurface and Seismic Ops | SLB-pth | SLB-pth | SLB-pth | SLB-pth |
| 2 | OE02 | QHSE PLAN: Surface and Injection Operations | ADM-ts | ADM-ts | ADM-ts | ADM-ts |
| 3 | OE03 | SITE SECURITY PLAN | ADM-mc | ADM-mc | ADM-mc | ADM-mc |
| 4 | OE04 | DRILLING PLANS, DEEP WELLS | SLB-pth | SLB-pth | SLB-pth | SLB-pth |
| 5 | OE05 | COMPLETION PLAN: IN-ZONE WELLS | SLB-jk | SLB-jk | SLB-jk | SLB-jk |
| 6 | OE06 | COMPLETION PLAN: GEOPHONE WELLS | SLB-ec | SLB-ec | SLB-ec | SLB-ec |
| 7 | OE07 | COMPLETION PLAN: SHALLOW gw MON WELLS | ISGS-rl | ISGS-rl | ISGS-rl | ISGS-rl |
| 8 | OE08 | CO2-RESISTANT WELL CONSTRUCTION - IN-ZONE WELLS | SLB-jk | SLB-jk | SLB-jk | SLB-jk |
| 9 | OE09 | INJECTION OPERATIONS ENVELOPE | TRIM-rm | TRIM-rm | TRIM-rm | TRIM-rm |
| 10 | OE10 | INJECTION OPS AND SHUTIN PLAN | TRIM-rm | TRIM-rm | TRIM-rm | TRIM-rm |
| 11 | OE11 | EQUIPMENT SPARING PLAN | TRIM-rm | TRIM-rm | TRIM-rm | TRIM-rm |
| 12 | OE12 | OPERATIONAL MONITORING PLAN | TRIM-rm | TRIM-rm | TRIM-rm | TRIM-rm |
| 13 | PM01 | PROGRAMMATIC AND OVERALL RISK | SLB-kht | SLB-kht | SLB-kht | SLB-kht |
| 14 | PM02 | PERMITS | ISGS-seg | ISGS-seg | ISGS-seg | ISGS-seg |
| 15 | PM03 | DATA MANAGEMENT PLAN | ISGS-db | ISGS-db | ISGS-db | ISGS-db |
| 16 | PM04 | BUDGET | ISGS-ab | ISGS-ab | ISGS-ab | ISGS-ab |
| 17 | PM05 | STAFF AND COORDINATION | ISGS-seg | ISGS-seg | ISGS-seg | ISGS-seg |
| 18 | PM08 | COMMUNICATIONS POLICY | ISGS-seg | ISGS-seg | ISGS-seg | ISGS-seg |
| 19 | PM12 | INJECTIVITY BACKUP PLAN | ISGS-sf | ISGS-sf | ISGS-sf | ISGS-sf |
| 20 | PM13 | CO2 MOVEMENT CONTINGENCY PLAN | SLB-jk | SLB-jk | SLB-jk | SLB-jk |
| 21 | PM14 | GOAL TRIAGE | ISGS-seg | ISGS-seg | ISGS-seg | ISGS-seg |
| 22 | PM15 | COMMUNICATIONS PLAN | ISGS-seg | ISGS-seg | ISGS-seg | ISGS-seg |
| 23 | SR01 | SITE SELECTION AND AOR | ISGS-rl | ISGS-rl | ISGS-rl | ISGS-rl |
| 24 | SR02 | USDW BASELINE CHEMISTRY | ISGS-rl | ISGS-rl | ISGS-rl | ISGS-rl |
| 25 | SR03 | RESERVOIR CHARACTERIZATION PLAN: IN-ZONE WELLS | SLB-ec | SLB-ec | SLB-ec | SLB-ec |
| 26 | SR06 | MVA PLAN | ISGS-rl | ISGS-rl | ISGS-rl | ISGS-rl |
| 27 | SR08 | SURFACE SEISMIC | SLB-mc | SLB-mc | SLB-mc | SLB-mc |
| 28 | SR09 | STATIC EARTH MODEL | ISGS-hl | ISGS-hl | ISGS-hl | ISGS-hl |
| 29 | SR10 | SIMULATION PLAN | SLB-aa | SLB-sm | ISGS-sf | SLB-os |

*Note: Certain Risk Response Action Groups (e.g. Geophone Wells) were pertinent only during the pre-injection project phase. Their RACI assignments are no longer active.*

| Abbreviations in RACI chart | |
|---|---|
| **ADM-mc** | ADM - Mark Carroll |
| **ADM-ts** | ADM - Tom Stone |
| **ISGS-ab** | ISGS - Alice Bernard |
| **ISGS-db** | ISGS - Dan Byers |
| **ISGS-hl** | ISGS - Hannes Leetaru |
| **ISGS-rl** | ISGS - Randy Locke |
| **ISGS-seg** | ISGS - Sallie Greenberg |
| **ISGS-sf** | ISGS - Scott Frailey |
| **SLB-aa** | Schlumberger Carbon Services - Ahsan Alvi |
| **SLB-ec** | Schlumberger Carbon Services - Ethan Chabora |
| **SLB-jk** | Schlumberger Carbon Services - Jim Kirksey |
| **SLB-kht** | Schlumberger Carbon Services - Ken Hnottavange-Telleen |
| **SLB-mc** | Schlumberger Carbon Services - Marcia Coueslan |
| **SLB-os** | Schlumberger Carbon Services - Ozgur Senel |
| **SLB-pth** | Schlumberger Carbon Services - PT Hughes |
| **SLB-sm** | Schlumberger Carbon Services - Scott Marsteller |
| **TRIM-rm** | Trimeric - Ray McKaskle |
| **OE** | **Operations and Execution RRAGs** |
| **PM** | **Project Management RRAGs** |
| **SR** | **Site Characterization and Monitoring RRAGs** |